AI for Good Sandbox Hackathon – Team MedVerse- MedVerse: Policy-Grounded Multi-Agent Institutional Coordination for Trustworthy Digital-Twin-Enabled Medical Metaverse Systems

Team leader: Aliyu Ibrahim | Members: Jeremiah Isuwa, Abubakar Isah, Kim Jinsul
Contact: ibal2010@yahoo.com | +82-1057250302 |

GitHub: https://github.com/ibal3233/MedVerse-Agent | Demo: https://drive.google.com/file/d/1i_TQl7wTeNfRoaE91CMIxkhKumJ3jZ8J/view

1. Introduction

Future Medical Metaverse systems will connect patients, digital twins, wearable and embodied sensors, immersive interfaces, AI decision support, hospitals, clinical-research platforms, and communication infrastructures. Different institutions govern these components, each with only partial information, different mandates, and different decision authority.

The resulting challenge is therefore not only prediction or diagnosis. It is institutional coordination under incomplete and asynchronously arriving information.

A healthcare provider may observe clinical deterioration, while a cybersecurity/network operator observes that the corresponding digital-twin telemetry is unreliable. A research institution may know that research consent has changed, while a privacy authority may determine that a proposed information exchange is not permissible. Each assessment can be locally correct while the overall system remains unable to determine a safe joint response.

MedVerse-Agent asks a simple but important question: when a patient's physical condition, digital twin, network, AI system, and regulatory state tell different stories, how should autonomous institutional agents coordinate a safe healthcare response?

2. Description of the use case

Scenario: A patient reports a fever via a medical-metaverse workspace alongside unverified telemetry and a concurrent research data request. Five independent agents (Hospital, Cyber, Privacy, Research/IRB, and Regulator) evaluate the event based strictly on their respective mandates.

Contribution: MedVerse-Agent acts as a policy-grounded multi-agent coordination layer bridging these institutional decisions and execution. By validating authority, harmonizing compatible actions, and recording auditable decision traces, it preserves institutional autonomy without forcing consensus or collapsing into a single authority. Figure 1 summarizes this architecture.

Scope: The prototype evaluates modeled Digital Twin workflows only. It demonstrates coordinated governance but explicitly does not connect to real systems, diagnose patients, prescribe treatments, or grant actual regulatory approvals.

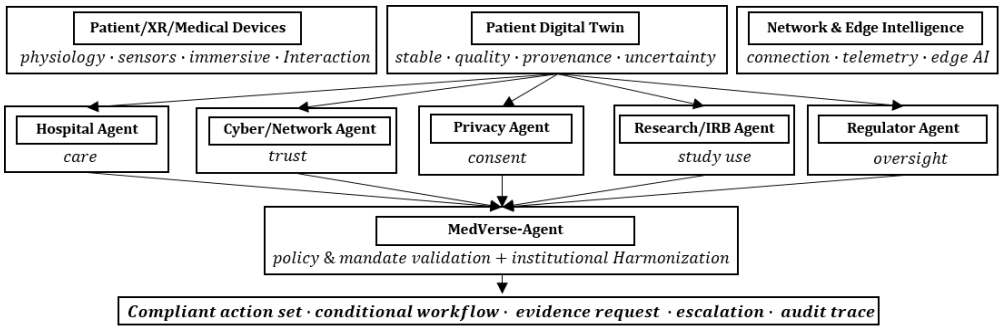


Figure 1. Medical-metaverse architecture: institution-specific agents act on role-scoped evidence while MedVerse-Agent coordinates policy-valid execution without acquiring their authority

3. Use case requirements

The MedVerse-Agent use case is driven by six core requirements: **R1** traceable, jurisdiction-eligible evidence; **R2** institution-owned observations and actions; **R3** provenance-aware Digital Twin (DT) state; **R4** feasible workflow dependencies and authorized alternatives; **R5** commitment decisions without future information; and **R6** reserved human authority with auditable decision receipts.

Evidence Architecture & Digital Twin: The system isolates three distinct evidence layers to prevent data contamination:

1. **Patient Digital Twin:** Records supplied scenario observations, provenance, and evidence quality states (VALID, DEGRADED, UNVERIFIED). These states reflect sensor and data reliability, not biological patient condition.
2. **Population Context:** Korean National Cancer Center (NCC) statistics provide optional oncology context but never supply patient-level facts.
3. **Korean Policy Corpus:** 44 curated records of official laws and guidelines ground institutional reasoning. The retrieval service applies jurisdiction and role-access restrictions before lexical (BM25) or semantic (MiniLM) ranking. Crucially, retrieved semantic relevance never independently authorizes an action. See Annex B. [7,9,12]

Institutional Boundaries & Contributions: Each agent operates strictly within its mandate, ensuring no single entity acts as a universal authority. Table 1 summarizes the roles and reserved boundaries.

Table 1. Institutional functions, policy anchors and reserved boundaries.

Agent	Policy function	Principal corpus sources	Boundary
Hospital / Healthcare Provider	Clinical care, record access, EMR integrity, permitted treatment use	Medical Service Act Arts. 21, 23; Health & Medical Data Utilization Guideline	May use clinical/DT evidence and allowed population context; must not take privacy, cyber or regulatory authority.
Cyber / Network	Telemetry trust, device/session integrity, safeguards, auditability	Medical Service Act Art. 23; PIPA Art. 28-4; PIPA Enforcement Decree Art. 29-5; Security Measures Standard; guideline safeguard sections	May assess network/security evidence and issue trust status; cannot change treatment or research authorization.
Privacy	Purpose limitation, consent scope, minimum necessary, disclosure, pseudonymization	PIPA Arts. 17, 18, 28-2 to 28-5; PIPA Enforcement Decree Art. 29-5; both official guidelines	Can permit/limit/deny data use or request authorization; cannot make clinical treatment decisions.
Research / IRB	Research authorization, consent, participant protection, IRB/DRB review	Bioethics and Safety Act Arts. 10, 16, 37; PIPA pseudonymization provisions; both official guidelines	Determines whether research processing is authorized and when IRB/DRB review or additional consent is required.
Regulator	Digital medical-product scope, medical-device trials, high-impact AI, human oversight	Digital Medical Products Act; DMPA Enforcement Rule; Medical Devices Act Art. 10; AI Basic Act Arts. 27, 32-35	May block or require review for high-impact automated action when evidence/governance conditions are not satisfied.

Reusability Scope: The coordination layer—including role-scoped routing, proposal contracts, policy validation, workflow harmonization, and auditable receipts—is fully reusable. Adapting this architecture to a non-medical domain requires supplying new authoritative sources, institutional mandates, and actions, rather than merely swapping the knowledge base.

Table 2. Source-to-action evidence and execution pathway.

Evidence construction and access	Institutional reasoning and execution
Official source → clause/section record → provenance/version metadata → eligibility → KR jurisdiction → agent access → semantic ranking → top-3 source-linked evidence.	Role-scoped observation → owned proposal → policy/mandate checks → harmonized workflow → COMMIT / bounded WAIT / permitted PREPARE → final revalidation → modeled effect and receipt.
Dense option: multilingual MiniLM-L12-v2, normalized 384-dimensional vectors; cosine-equivalent dot product. BM25 is an explicitly separate lexical backend. Exact-ID lookup bypasses ranking, not eligibility/access.	The live workflow uses deterministic institution-owned contracts. Learned Privacy/Cyber proposal components were evaluated separately. The coordinator and reference evaluator are not extra institutional agents.

As summarized in Table 2, the source-to-action pathway separates evidence construction and access from institutional reasoning and execution. Evidence retrieval and workflow validation remain distinct: semantic relevance does not establish applicability or authorization. Detailed corpus and Digital Twin evidence are provided in Annex B and Annex F/J.

4. Mapping with ITU-T Y.3172, AI Readiness 2.0 and domain-specific policies

Y.3172 supplies the functional ML architecture; AI Readiness Report 2.0 supplies complementary assessment dimensions. Table 3 adapts the official template's use-case / readiness / documents columns to this implementation. Detailed readiness evidence is provided in Annex B6.

Table 3. Y.3172, AI Readiness 2.0 and policy-evidence mapping.

Node	Medical-metaverse function and evidence	Readiness 2.0 link	Documents and applicability boundary
SRC	R1/R3: official Korean records plus modeled clinical, network, consent and research events. Inspect source IDs and observation timestamps.	D1 / D4: metadata-bearing local evidence; scenario provenance is explicit.	Medical Service Act (MSA); PIPA; Bioethics and Safety Act (BSA); DMPA and AI Basic Act candidate corpus. [7]
C	R1/R2: capture source versions and release each event only to its permitted institutional view.	D1 / D4: traceable collection and local context. No patient data collection is claimed.	MSA Arts. 21, 23; PIPA purpose/access and security provisions; modeled collection permissions. [7]
PP	R1/R3: clause/section records, labelled bilingual representations, role metadata and per-field DT provenance.	D1 / D5: reproducible representation. TRAIN-fitted scaling/missingness belongs to the separate Cyber study.	PIPA safeguards; Health & Medical Data Utilization and Pseudonymized Information Processing guidance. [7]

M	R1/R2: dense/BM25 retrieval and institutional proposals; separate learned-agent assessment. Show backend, source rank and per-class results.	D1 / D4 / D5: model provenance, language-specific retrieval and component evaluation.	Source/model registry; MSA/PIPA/BSA evidence relevant to the particular activity. Model output is not legal authority. [7,8]
P	R2/R4/R5: action-owner, purpose, scope, evidence and current-authority checks; bounded timing and precommit revalidation.	D7 / D8 / D10: align obligations, retain review authority and inspect policy consequences.	MSA; PIPA; BSA review/consent provisions; scoped oversight sources. Timing limits are modeled contracts, not statutory clinical deadlines. [7]
D	R4/R6: identify permitted recipients and distribute approved workflow outputs, handoffs and minimum-necessary views.	D5 / D7: observable handoffs and controlled distribution. Energy/deployment gains are not measured.	PIPA disclosure/safeguard anchors; institutional access/audit procedures remain review-pending. [7]
SINK	R5/R6: operator workspace, conditional holds, human escalation and modeled side effects with exportable receipts.	D6 / D8 / D10: inspectable interactions and policy sandbox; no human usability study claimed.	Applicable institutional SOPs and reserved human authority; WHO/OECD are comparative guidance, not Korean permission. [5–7]

Dimension labels: D1 Data/model Marketplace; D4 Contextualization and Regional Impact; D5 Level of Integration of AI in Workflows; D6 Human Interface; D7 Strategy Alignment; D8 Collaboration with AI; D10 AI & Policies. [2]

The harmonizer coordinates across the functional pathway; it is not renamed as the Y.3172 distributor. D distributes outputs to their sinks. The diagram's arrows are not evidence that one unrestricted LLM owns every stage.

Implemented evidence versus remaining readiness work

The strongest evidence concerns D7, D8 and D10: distinct institutional obligations, inspectable modeled reviews and action-policy tests. Reuse is a D3 design opportunity; bilingual retrieval contributes to D4/D6 but does not prove arbitrary-language access. D12 energy and D13 deployment infrastructure remain unmeasured beyond a local runtime. No generative-content marketplace, cloud-hosted ITU integration or national adoption impact is claimed. Figure 2 shows the source-controlled policy pathway.

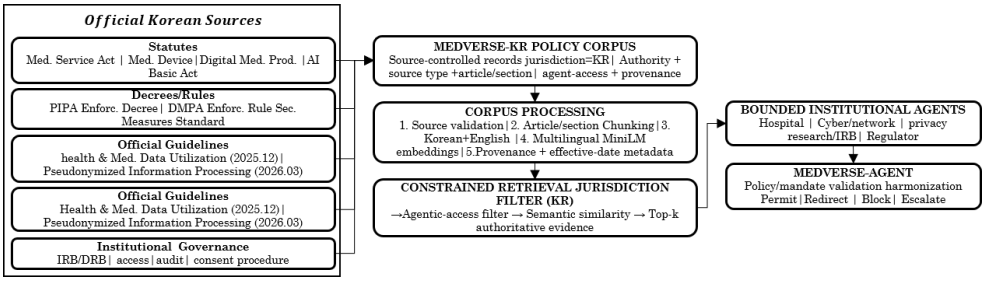


Figure 2. Source-controlled Korean policy corpus and its role in MedVerse-Agent.

5. Error scenarios

The prototype is evaluated through a progressive oncology scenario in which each step exposes a coordination risk addressed by MedVerse-Agent.

Scenario 1 – Clinical concern under uncertain telemetry. A chemotherapy patient reports fever while telemetry becomes stale or unverifiable. Clinical review must continue, requiring the system to distinguish patient condition from evidence quality.

Scenario 2 – Policy-grounded institutional disagreement. The Hospital proceeds with clinical review while Cyber/Network marks affected telemetry unverified and a research-access request arrives. These locally valid but different actions motivate the B1–B5 architectural comparison and institutional harmonization.

Scenario 3 – Conditional research disclosure. Privacy and Research/IRB evaluate the request's purpose, scope, authority, and supporting evidence. This tests whether the system respects institutional boundaries and prevents unauthorized disclosure.

Scenario 4 – Partial information and commitment timing. New alternatives or authority conditions may arrive after an action becomes available. Committing too early can block a later authorized workflow, while waiting may forfeit an earlier opportunity. This motivates comparison of eager commitment, fair bounded wait, and MedVerse commitment.

At each stage, MedVerse-Agent records released observations, retrieved evidence, institutional proposals, authority checks, coordination decisions, final modeled actions, and escalations. It is a policy-grounded multi-agent coordination layer, not a diagnostic or treatment engine.

Each stage generates an auditable trace containing observation provenance, retrieved evidence, institutional proposal, authority checks, coordination state and final modeled action or escalation. The trace records decision rationale, not private model chain-of-thought. The evaluation tests whether explicit institutional coordination improves policy-compliant workflow behavior rather than clinical prediction accuracy.

The progressive oncology scenario releases a clinical concern, technical-integrity evidence, a research request, DT context, institution-owned alternatives, and modeled authority in stages. We evaluate the concept through three questions: does explicit institutional coordination improve the joint workflow? are institutional proposals grounded in reliable evidence? under partial information, when should a valid proposal be committed, held, prepared or escalated? Detailed component studies are mapped in Annex A2 and reported in Annex C–G. The major findings are as follows.

Evaluation Framework and Research Questions

The questions in Table 6. consolidate the completed studies for presentation; they are not a retrospective claim of a single preregistered benchmark. B1–B5 remain the original architecture comparison. Later methods vary one internal mechanism and are not new institutional agents. Collectively, the results show that MedVerse-Agent supports evidence-grounded, authority-aware coordination and inspectable execution, while retaining strong-comparator ties, language sensitivity, and the benefit–cost trade-off of waiting.

Table 5 Evaluation questions and principal findings. Detailed protocols, figures, denominators and evidence identifiers are provided in Annex A–G.

Question	Comparison unit	Main finding and implication
RQ1 · Institutional assessment	Privacy repair / Cyber prospective study	Cyber action and assessment: 72/72 correct; Privacy action: 486/486. Bounded modeled decision spaces; not live LLM performance.
RQ2 · Evidence validity and language	DT adapter; BM25 versus multilingual embeddings	Known-query Recall@3: BM25 EN/KO 85%/85%; dense 95%/75%. Freshness remains field-specific; language sensitivity remains.
RQ3 · Architecture	Frozen B1–B5 reference architectures	B4→B5 coordination quality .600→1.000 and exact match .143→1.000 on seven archetypes / 70 variants.
RQ4 · Workflow mechanism	Scheduling variants; alternative selectors	Fair retry and dependency ordering both 318/318. Three capable selectors complete 10/10 feasible snapshots; no unique superiority.
RQ5 · Dynamic commitment	Eager, fair bounded wait, MedVerse commitment	14/32 versus 15/32 complete; four gains and three losses. Fair control ties. Original C01/C02 remain unchanged.
RQ6 · Integrated behavior	Direct engine versus HTTP/UI replay	720/720 event states and 120/120 traces agree. Per-role effects are observable; no patient, security or clinical efficacy claim.

Architectural comparison

B1 = centralized/omniscient reference; B2 = unconstrained role-based MAS; B3 = consensus/voting reference; B4 = independent policy filtering; B5 = MedVerse-Agent harmonization. B3 implements majority-intent voting, not a learned multi-round debate. [9] Table 5 reports the B1-B5 results.

Table 4. Original 70-case architectural comparison. IC = institutional correctness; CQ = coordination quality; EWM = exact workflow match; PVR = policy/authority violation rate; UER = unnecessary escalation rate; IOR = information overexposure rate; UXR = unsafe execution rate. Exact definitions and denominators are in Annex A3.

Method	IC ↑	CQ ↑	EWM ↑	PVR ↓	UER ↓	IOR ↓	UXR ↓
B1	.179	.393	.000	1.000	.000	.929	1.000
B2	.821	.540	.000	.143	.000	.929	.143
B3	.607	.286	.000	.000	.667	.929	.000
B4	.964	.600	.143	.000	.333	.000	.000
B5	1.000	1.000	1.000	.000	.000	.000	.000

Dynamic coordination under partial information

Later studies isolate scheduling, authorized alternative selection and commitment timing rather than redefining B1–B5. Dependency-aware ordering improves order robustness over single-pass processing but ties a fair stateful retry control; capable workflow selectors also tie on the finite alternative library. The final dynamic study therefore asks when an authorized action should be committed, held, prepared or escalated when future proposals are unknown. Detailed mechanism evidence is in Annex E–G. Figure 3 shows the commitment-timing trade-off.

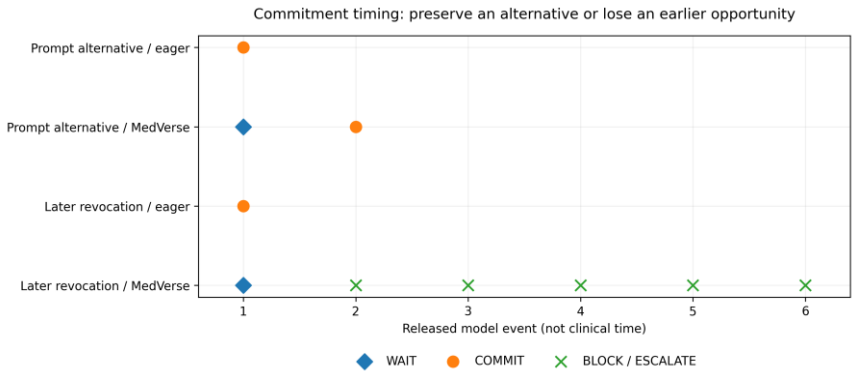


Figure 3. Commitment timing in the live workspace. P01 shows a benefit of waiting; P04 shows its opportunity cost. Event steps are engineering assumptions, not clinical time.

As quantified in Table 6, waiting can preserve an authorized alternative but can also lose an earlier valid opportunity; MedVerse gains four continuations and loses three versus eager commitment while tying the capable bounded-wait control

*Table 6. Bounded commitment under partial information. *Conditional on a feasible modeled clinical obligation. Eight prefix families, four related continuations each; the capable bounded-wait control ties MedVerse. Timing and authority are separate: PREPARE is not care, disclosure or approval.*

Timing controller	Complete	Clinical on time*	Wait steps	Invalid commits
Eager commitment	14/32	28/30	0	0
Fair bounded wait	15/32	29/30	38	0
MedVerse commitment	15/32	29/30	38	0

6. Policy and standards gaps derived from the mapping exercise

The mapping indicates that existing ITU work addresses ML pipelines, autonomous-network trust, Digital Twin architecture and metaverse interoperability, while MedVerse-Agent exposes additional requirements for coordinating institution-owned decisions across these layers. Table 7 summarizes the resulting standardization opportunities; the detailed requirement-to-standard mapping is provided in Annex H2.

Table 7. Standards mapping and proposed coordination opportunities.

ITU area	Opportunity identified
Y.3172 [1]	Add machine-readable institutional mandate, jurisdiction and action-authority metadata between model output and execution
Y.3060/Y.3061 [11],[12]	Extend trust to evidence provenance and institution-owned authority across multiple controllers
Y.3090/Y.4239/Y.4240 [13],[3],[4]	Healthcare DT exchange should preserve freshness, provenance, purpose, recipient and authority context
FG-AI4H / FG-MV [15],[16]	Runtime institutional validation, minimum-necessary information exchange and auditable coordination

Impact: MedVerse-Agent makes institutional disagreement operationally inspectable, preserves permitted work while holding unauthorized actions, and exposes when early commitment can block later authorized workflows.

References

[1] ITU-T Y.3172 (2019), Architectural framework for machine learning in future networks including IMT-2020.Official source

[2] ITU, AI Ready — Analysis Towards a Standardized Readiness Framework, Report 2.0, January 2026; §3 Dimensions.Official source

[3] ITU-T Y.4239 (2025), Reference model for integrating virtual and physical worlds through digital twins in the metaverse.Official source

[4] ITU-T Y.4240 (2025), Interoperability for integrating virtual and physical worlds through digital twins in the metaverse.Official source

[5] WHO, Ethics and governance of artificial intelligence for health (2021).Official source

[6] OECD, Recommendation of the Council on Health Data Governance, OECD/LEGAL/0433.Official source

[7] Team MedVerse candidate Korean corpus v0.4 and source snapshots. Official Korean originals are the policy anchors; interpretation review remains pending.

[8] Sentence Transformers, paraphrase-multilingual-MiniLM-L12-v2 model card.Official source

[9] Team MedVerse frozen Gates 22B.6–33, supplied local console results, status files and source implementations; identifiers listed in I1.

[10] AI for Good Sandbox Hackathon submission template, supplied by the user; UAE competition requirements.Official source

Official source[11] ITU-T Y.3060 (2023) — Autonomous networks: Overview on trust.

[12] ITU-T Y.3061 (2023) — Autonomous networks: Architecture framework.

[13] ITU-T Y.3090 (2022) — Digital twin network: Requirements and architecture.

[14] National Cancer Center, Republic of Korea — Cancer Incidence Statistics (Public Data Portal, 2026).

[15] ITU-T Focus Group on Metaverse (FG-MV) — approved deliverables on metaverse architecture, cyber risk, PII and interoperability.

[16] ITU/WHO Focus Group on AI for Health (FG-AI4H) — deliverables on ethics, regulation, data, evaluation and AI lifecycle.

Full references, evidence identifiers and reproducibility locations are provided in Annex I1–I2.

Supporting Evidence Annex

This annex supports the five-page main report with detailed policy-corpus documentation, metric definitions, component validation, architecture and coordination studies, Digital Twin/retrieval evidence, standards mapping and reproducibility records. It introduces no new experiments and does not modify any reported result, threshold or comparator.

A1. Evaluation study taxonomy

Table A1 distinguishes the study families, varied components and comparator interpretations.

Table A1. Evaluation study taxonomy.

Study family	What varies	Actual comparator names / interpretation
Institutional learners	Learned assessment/action heads	Privacy action-head repair; Cyber action/assessment multinomial logistic regression. C is a hyperparameter, not an agent or an epoch.
Policy retrieval	Evidence-ranking backend	BM25 lexical versus multilingual dense embeddings. Filters and target labels are held explicit.
Architecture (B1–B5)	Whole reference architecture	B1 omniscient; B2 unconstrained roles; B3 voting/debate-labelled reference; B4 policy filter only; B5 MedVerse.
Scheduling	Order/revisiting of actual proposals	SNAPSHOT_ONCE; SEQUENTIAL_ONCE; STATEFUL_FIXED_POINT; DEPENDENCY_ORDERED; DEPENDENCY_EDGES_REMOVED.
Workflow selection	Selection among institution-owned alternatives	REQUEST_ONLY; FAIR_STATEFUL_SELECTOR; INDEPENDENT_CONSTRAINT_PLANNER; MEDVERSE_HARMONIZER.
Commitment timing	When permitted effects commit	Eager commitment; fair bounded wait; MedVerse commitment. The full-information reference is noncausal and evaluation-only.
Integration fidelity	Execution path / observation adapter / UI wrapper	Direct engine versus HTTP/UI behavior; old versus new observation diagnostics. No new controller is inferred from a UI label.

The five institutions are constant roles, not the five B configurations, and a timing controller is not an additional agent. The policy-filter baseline and the later fair stateful selector are distinct implementations with different tasks and denominators.

A2. Evidence map supporting the three main research questions

Table A2 maps the three main research questions to their supporting evidence.

Table A2. Detailed evidence map supporting the three main research questions.

RQ	Question	Evidence
RQ1	Are institutional proposals grounded in reliable evidence?	Cyber/Network and Privacy validation; Digital Twin provenance; Korean/English retrieval (Annex C and F).
RQ2	Does explicit multi-agent institutional coordination improve the joint workflow?	B1–B5 architecture; proposal scheduling; authorized workflow selection; progressive execution (Annex D–E).
RQ3	Under partial information, when should a valid proposal be committed, held, prepared or escalated?	Bounded commitment, institutional contribution and interface replay (Annex G).

The three-question structure is a presentation-level synthesis. The individual study protocols and evidence records define what was specified before execution.

A3. Metric definitions, denominators and interpretation limits

The original seven architectural metrics retain their frozen operational definitions. Table A3 summarizes the evaluation implementation; no scoring rule is changed. The implementation and evidence identifier are listed in Annex I1.

Table A3. Metric definitions and interpretation limits

Configuration	Description
B1: Single Omniscient Agent	Centralized agent with access to all institutional information.
B2: Unconstrained Multi-Agent	Role-based agents without enforced institutional observation, action, or communication boundaries.
B3: Consensus/Debate MAS	Institutional agents attempt to resolve their decisions through consensus or debate.
B4: Policy-Filter-Only MAS	Independent institutional policy enforcement without cross-agent workflow harmonization.

B5: MedVerse-Agent	Mandate-aware agents with constrained policy retrieval, validated actions, institutional harmonization, and auditability.
Metric	**Definition / Purpose**
Institutional Correctness (IC)	Fraction of institutional decisions consistent with expected actions and authority boundaries.
Policy Violation Rate (PVR)	Fraction of evaluated actions violating configured policy or mandate constraints.
Coordination Quality (CQ)	Correctness of joint institutional decisions and workflow components under the frozen scoring protocol.
Exact Workflow Match (EWM)	Fraction of scenarios matching the complete expected institutional workflow.
Unnecessary Escalation Rate (UER)	Fraction of eligible scenarios escalated when a compliant workflow was feasible.
Information Overexposure Rate (IOR)	Fraction of shared information exceeding the defined institutional information requirement.
Unsafe Execution Rate (UXR)	Fraction of executed actions violating the benchmark's admissibility conditions.

B1. Korean policy corpus, institutional scope and source links

The following tables present the Korean policy knowledge base underpinning MedVerse-Agent, identifying relevant laws, regulations, and official guidelines, together with their applicability to the five institutional agents. They illustrate how policy evidence is organized, mapped to institutional mandates, and used to support jurisdiction-aware retrieval, policy validation, and trustworthy coordination within the medical metaverse Table B1 summarizes the corpus layers; Table B2 maps policy functions to sources and agents; Tables B3-B5 report source inventory and institutional scope; and Table B6 summarizes the AI Readiness evidence status.

Table B1. Canonical Korean policy-corpus structure.

Corpus layer	**Included sources**	**Primary system function**	**Primary agents**
Statutes	Medical Service Act; PIPA; Bioethics and Safety Act; Medical Devices Act; Digital Medical Products Act; AI Basic Act	Define legal authority, prohibited/allowed processing, consent, clinical/research obligations, high-impact AI duties	Hospital; Privacy; Research/IRB; Regulator; Cyber where security/integrity applies
Decrees / Enforcement Rules	PIPA Enforcement Decree; Digital Medical Products Enforcement Rule; Personal Information Security Measures Standard	Translate statutory duties into safeguards, access separation, records, implementation and safety requirements	Privacy; Cyber/Network; Research/IRB; Regulator
Official Guidelines	Health & Medical Data Utilization Guideline (2025.12); Pseudonymized Information Processing Guideline (2026.03)	Operationalize pseudonymization, risk review, data sharing, AI research use, IRB/DRB, safe processing environments and documentation	Privacy; Research/IRB; Hospital; Cyber/Network; Regulator where AI governance intersects
Institutional Governance	IRB/DRB procedures; access, audit, consent and security procedures	Determine who must review, what evidence must be recorded, and when MedVerse-Agent must redirect or escalate	All agents through mandate and workflow constraints

Table B2. Functional mapping: what the corpus enables.

Function in MedVerse-Agent	**Policy source**	**Agent(s)**	**Operational use**	**Official source**
Patient-record access and disclosure boundary	Medical Service Act Art. 21	Hospital Agent	Controls access/disclosure of medical records and supports minimum-authority routing.	Open official source
EMR integrity, preservation and audit logs	Medical Service Act Art. 23	Hospital + Cyber/Network	Grounds EMR safe management, anti-	Open official source

			alteration/leakage, and access-log preservation.	
Third-party disclosure and purpose limitation	PIPA Arts. 17-18	Privacy Agent	Determines whether disclosure/use is within authorized purpose, consent and legal basis.	Open official source
Pseudonymized research use	PIPA Art. 28-2	Privacy + Research/IRB	Permits bounded secondary use for statistics/scientific research/public-interest archiving and constrains third-party provision.	Open official source
Cross-controller pseudonymized-data combination	PIPA Art. 28-3	Privacy + Research/IRB	Requires specialized combination mechanisms and controlled export of combined information.	Open official source
Safeguards and processing records	PIPA Art. 28-4 + Enforcement Decree Art. 29-5	Privacy + Cyber + Research/IRB	Grounds separation of additional information, access control, safeguards, processing records and retention.	Open official source
Re-identification prohibition / stop-and-destroy response	PIPA Art. 28-5	Privacy + Research/IRB	Blocks deliberate re-identification and supports stop/recovery/destruction actions if identity is exposed.	Open official source
IRB authority and research oversight	Bioethics and Safety Act Art. 10	Research/IRB Agent	Defines institutional IRB authority for ethical/scientific review, consent and participant protection.	Open official source
Human-subject consent	Bioethics and Safety Act Art. 16	Research/IRB Agent	Grounds consent requirements for human-subject research.	Open official source
Research consent withdrawal / human-derived material	Bioethics and Safety Act Art. 37	Research/IRB Agent	Supports withdrawal-aware research restrictions and audit/evidence handling.	Open official source
Medical-device clinical trial approval and participant consent	Medical Devices Act Art. 10	Regulator + Research/IRB	Constrains investigational medical-device trials and participant protection.	Open official source
Digital medical-product scope	Digital Medical Products Act Art. 2	Regulator Agent	Determines whether a digital product falls within the regulated digital-medical-product scope.	Open official source
Digital medical-product safety evidence	DMPA Enforcement Rule Art. 14	Regulator Agent	Supports safety/validation/cybersecurity evidence requirements used by the regulator agent.	Open official source
High-impact AI safety, risk management and human oversight	AI Basic Act Arts. 27, 32-35	Regulator Agent	Grounds high-impact classification, safety/reliability, human management/oversight, documentation and impact assessment.	Open official source
Operational health-data governance	Health & Medical Data Utilization Guideline (2025.12)	Privacy + Research/IRB + Hospital + Cyber	Operationalizes health-data pseudonymization, minimum necessary processing, IRB/DRB coordination, risk review and controlled AI-research environments.	Open official source
Risk-based pseudonymization and AI-development workflows	Pseudonymized Information Processing Guideline (2026.03)	Privacy + Research/IRB + Cyber; Regulator where AI intersects	Defines the five-stage pseudonymization workflow, low/medium/high risk model, safe management, non-structured data treatment, AI-development examples and required records.	Open official source
Technical/administrative/physical security controls	Personal Information Security Measures Standard	Cyber/Network Agent	Grounds minimum safeguards, access controls and security-management expectations for personal information systems.	Open official source

Inventory reproduced from the preserved candidate corpus. The 44 records are indexed policy entries, not 44 independent statutes. Thirteen document identifiers include two Medical Service Act identifiers. Forty-two records are retrieval-eligible;

two informational press-release records are excluded. Internal verification labels describe project provenance, not certified applicability. Table B3 reports the document-group inventory.

Table B3. Corpus inventory by document group.

Source group / document identifier grouping	Records	Eligible	Type	Recorded date
Medical Service Act	1	1	statute	2026-10-02
Personal Information Security Measures Standard	1	1	administrative_rule	2026-07-01
Personal Information Protection Act	6	6	statute	2026-09-11
Bioethics and Safety Act	3	3	statute	2025-10-01
Digital Medical Products Act	1	1	statute	2026-10-08
Enforcement Rule of the Digital Medical Products Act	1	1	enforcement_rule	2026-01-24
Medical Service Act	1	1	statute	2026-10-02
Enforcement Decree of the Personal Information Protection Act	1	1	decree	2026-09-11
Medical Devices Act	1	1	statute	2026-10-08
Framework Act on the Development of Artificial Intelligence and Establishment of Trust	5	5	statute	2026-07-21
Health and Medical Data Utilization Guideline	10	10	official_guideline	Not recorded
PIPC Press Release on 2026 Pseudonymized Information Guideline Revision	2	0	press_release	Not recorded
Pseudonymized Information Processing Guideline	11	11	official_guideline	Not recorded

Table B4 summarizes the evidence records eligible for each institutional role.

Table B4. Role-eligible evidence scope.

Role	Eligible records	Meaning
Hospital	6	Shared records may serve several roles; not a completeness or accuracy score.
Cyber/Network	12	Shared records may serve several roles; not a completeness or accuracy score.
Privacy	28	Shared records may serve several roles; not a completeness or accuracy score.
Research/IRB	30	Shared records may serve several roles; not a completeness or accuracy score.
Regulator	8	Shared records may serve several roles; not a completeness or accuracy score.

B2. Active policy anchors and readiness evidence status

Table B5 maps the active policy anchors to each institution and states the corresponding reserved boundary.

Table B5. Corpus-to-agent allocation and reserved boundaries.

Agent	Policy function	Principal corpus sources	Boundary
Hospital / Healthcare Provider	Clinical care, record access, EMR integrity, permitted treatment use	Medical Service Act Arts. 21, 23; Health & Medical Data Utilization Guideline	May use clinical/DT evidence and allowed population context; must not take privacy, cyber or regulatory authority.
Cyber / Network	Telemetry trust, device/session integrity, safeguards, auditability	Medical Service Act Art. 23; PIPA Art. 28-4; PIPA Enforcement Decree Art. 29-5; Security Measures Standard; guideline safeguard sections	May assess network/security evidence and issue trust status; cannot change treatment or research authorization.
Privacy	Purpose limitation, consent scope, minimum necessary, disclosure, pseudonymization	PIPA Arts. 17, 18, 28-2 to 28-5; PIPA Enforcement Decree Art. 29-5; both official guidelines	Can permit/limit/deny data use or request authorization; cannot make clinical treatment decisions.
Research / IRB	Research authorization, consent, participant protection, IRB/DRB review	Bioethics and Safety Act Arts. 10, 16, 37; PIPA pseudonymization provisions; both official guidelines	Determines whether research processing is authorized and when IRB/DRB review or additional consent is required.
Regulator	Digital medical-product scope, medical-device trials, high-impact AI, human oversight	Digital Medical Products Act; DMPA Enforcement Rule; Medical Devices Act Art. 10; AI Basic Act Arts. 27, 32-35	May block or require review for high-impact automated action when evidence/governance conditions are not satisfied.

Readiness mapping is evidence-specific, not a checklist of claimed dimensions

Table B6 summarizes the directly evidenced, design-level and unmeasured AI Readiness dimensions.

Table B6. AI Readiness evidence status.

Evidence status	Relevant dimensions	Interpretation
Direct prototype evidence	D1, D4, D5, D6, D7, D8, D10	Corpus/model metadata; KR context; workflow implementation; inspectable interface; institutional alignment; modeled review and policy experiments.
Design-level extension	D3 Cross-domain correlation analysis	Healthcare-to-education role/configuration mapping. No transfer performance or education deployment is claimed.
Not measured as impact	D9, D11, D12, D13	No expert-benefit study, population inclusion benefit, energy saving or field infrastructure-readiness score.
Not claimed	D2 Generated Content Marketplace	This release is not a generative content marketplace; a sentence embedding is not an LLM explanation service.

C1. Cyber/Network prospective validation

Cyber action and assessment accuracy/macro-F1 are 1.000 on 72 modeled prospective records, four groups of 18 (g5–g8). This is a historical learned-agent component study, separate from the later live deterministic contracts. The class supports are visible in the confusion matrix. [9] Figure C1 reports class-level action correctness, and Figure C2 shows stability across the four prospective groups.

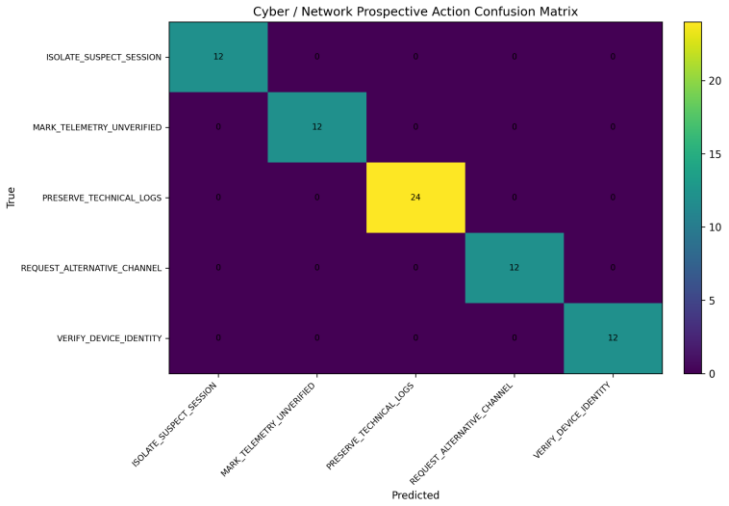


Figure C1. Action confusion matrix: 12, 12, 24, 12 and 12 correct classifications. Total 72; no off-diagonal counts in the supplied plot.

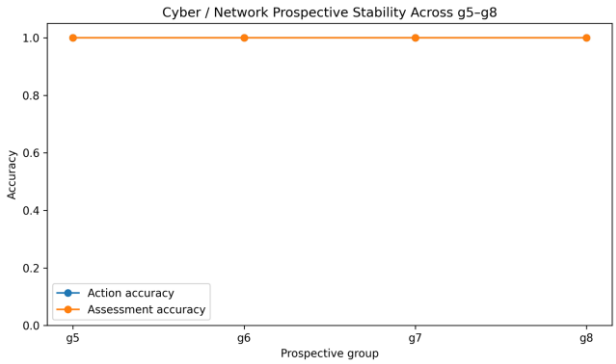


Figure C2. Group action/assessment accuracy both equal 1.000; the curves overlap. Stability is across the four modeled prospective groups, not external networks.

Clinical implication is bounded: the component can reproduce its defined institutional action labels under the tested shift. It is not evidence that Cyber can diagnose a patient or that every future attack is detectable.

C2. Model selection and Privacy-agent validation

Figure C3 shows the development-only hyperparameter sweep, and Figure C4 reports the corresponding LBFGS solver iterations.

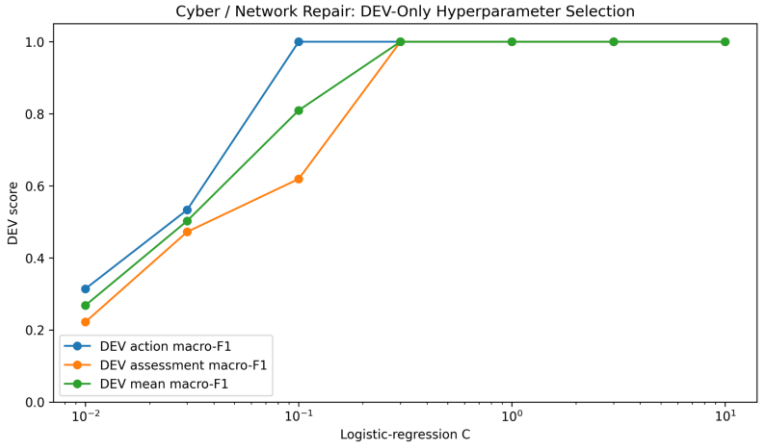


Figure C3. DEV-only C sweep. The documented tie-break selects C = 0.3 before prospective evaluation. These points are hyperparameter settings, not epochs.

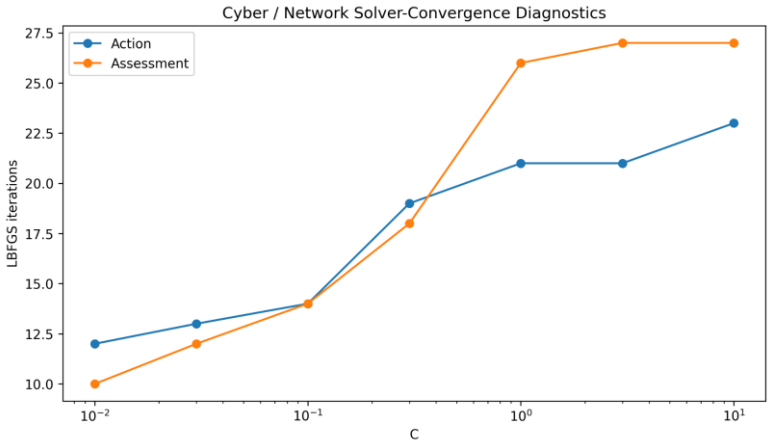


Figure C4. LBFGS iteration counts by C for the action and assessment heads. This is solver effort across fitted models, not a loss-versus-epoch training curve.

Cyber uses 54 TRAIN and 18 DEV records. TRAIN-fitted standardization and explicit missingness address the diagnosed representation issues; opened g4 diagnostic records are excluded from the prospective test. The model family remains multinomial logistic regression. [9] Table C1 summarizes the confirmed component results and interpretation boundaries.

Table C1. Institutional component validation summary.

Historical component	Confirmed result	Boundary
Privacy action head	486/486 action decisions; macro-F1 and minimum action recall 1.000; unsafe-ALLOW execution rate 0.	Fresh contexts in an already-defined finite policy space. Some action classes have support 3.
Privacy evidence head	Accuracy 0.963; macro-F1 0.945.	Not every head is perfect; non-action heads were not repaired with the action head.
Research/IRB learning	Exploratory/deferred.	Do not promote exploratory evidence to the Privacy/Cyber confirmation status.

Source: supplied five-page report and preserved Gate22B.6/Gate24C status files. Historical learned components are not silently substituted for the deterministic live proposal library.

C3. Class-level performance and training-support analysis

The following tables retain the original learned-component evaluation. A confusion matrix, a DEV hyperparameter sweep, a training-support history and a solver-iteration count answer different questions; none is an epoch-by-epoch learning curve. These measurements are not substituted for live coordination performance. [9] Table C2 gives the class-level Cyber metrics, while Table C3 summarizes performance as TRAIN support grows.

Prospective Cyber action classification

Table C2. Prospective Cyber action classification.

Action class	Precision	Recall	F1	Support
Isolate Suspect Session	1.000	1.000	1.000	12
Mark Telemetry Unverified	1.000	1.000	1.000	12
Preserve Technical Logs	1.000	1.000	1.000	24
Request Alternative Channel	1.000	1.000	1.000	12
Verify Device Identity	1.000	1.000	1.000	12

Source: archived gate24c_status.json. Four prospective groups g5–g8; 72 records in total. The action and assessment heads both achieve aggregate accuracy/macro-F1 1.000 within this modeled generator shift.

Cumulative TRAIN support at the selected C = 0.3

Table C3. Cyber training-support history.

Support step	TRAIN records	DEV action F1	DEV assessment F1	Action iterations	Assessment iterations
1	18	1.000	0.619	14	14
2	36	1.000	1.000	17	18
3	54	1.000	1.000	19	18

The archived sequence adds groups g0, g1 and g2: 18 → 36 → 54 TRAIN records, with 18 DEV records held for model selection. Assessment DEV macro-F1 rises from 0.619 to 1.000 as support grows; action DEV macro-F1 is 1.000 at all three recorded support steps. This is descriptive training-support history—not evidence that the prospective groups participated in selection.

What these results contribute to the submission

They demonstrate a separately evaluated institutional proposal component and a disciplined response to representation failure. The strongest methodological point is that the repair changed TRAIN-fitted scaling and explicit missingness rather than changing the policy boundary or concealing the initial failure. Prospective labels and support remain fixed.

Privacy evidence is summarized in Annex C2. Its action result is 486/486 within the finite predefined state space, while the evidence head remains imperfect. Research/IRB learning remains exploratory. No learned-action score should be used as a claim that an action is legally authorized, that all five live roles are learned models, or that unseen clinical situations have been validated.

Source files: gate24c_status.json; gate24c_training_history.csv; gate24c_hyperparameter_results.csv; supplied Cyber figures. No fitting or evaluation was repeated for this editorial revision.

D1. Architectural comparison: B1–B5

The architectural study evaluates five frozen coordination configurations on the same 70 benchmark records spanning seven coordination archetypes with ten context variants each. No baseline-specific tuning is introduced. [9] Figure D1 presents the coordination outcomes, and Figure D2 reports the safety and governance metrics.

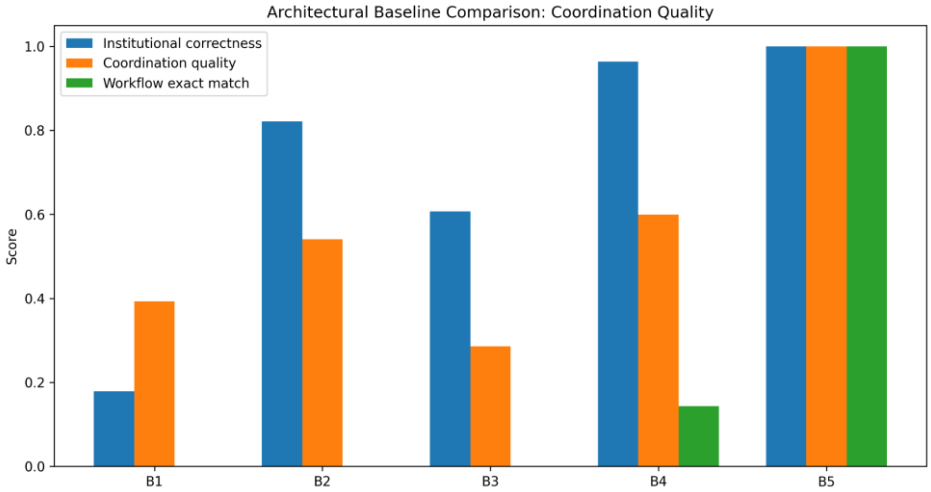


Figure D1. Original IC/CQ/EWM results. The full B5 reference realizes the encoded workflow; this is not a comparison of optimized general-purpose planning systems.

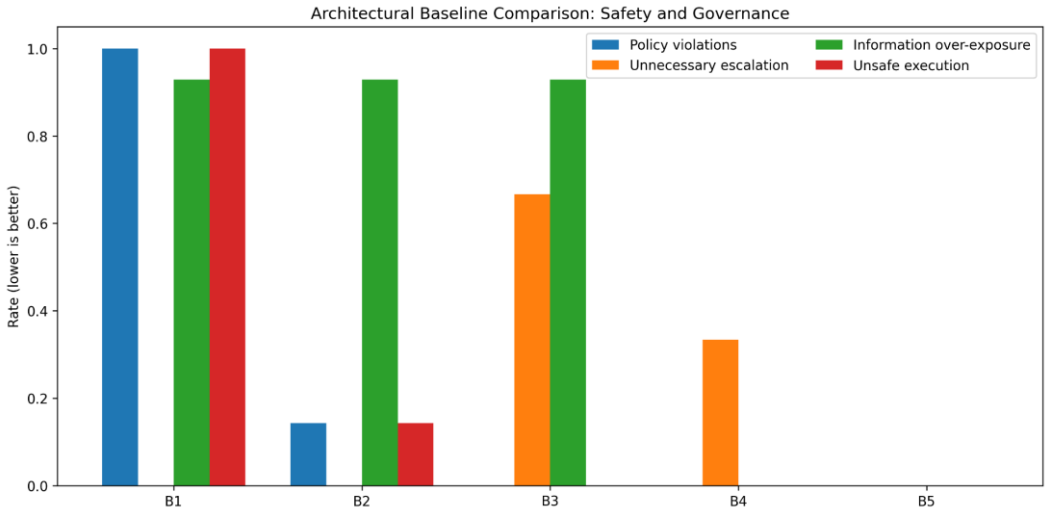


Figure D2. Original policy/authority, overexposure, escalation and unsafe-token metrics. Zero unauthorized actions must be interpreted alongside useful completion.

The strongest bounded observation is B4→B5: individual filtering alone does not reproduce the encoded cross-institutional workflow. Later studies test the mechanism with stronger stateful controls rather than assuming this result proves general superiority.

D2. Baseline architecture definitions and archetype coverage

Figure D3 reports exact workflow matching by coordination archetype, and Table D1 defines the five baseline architectures and their interpretation limits.

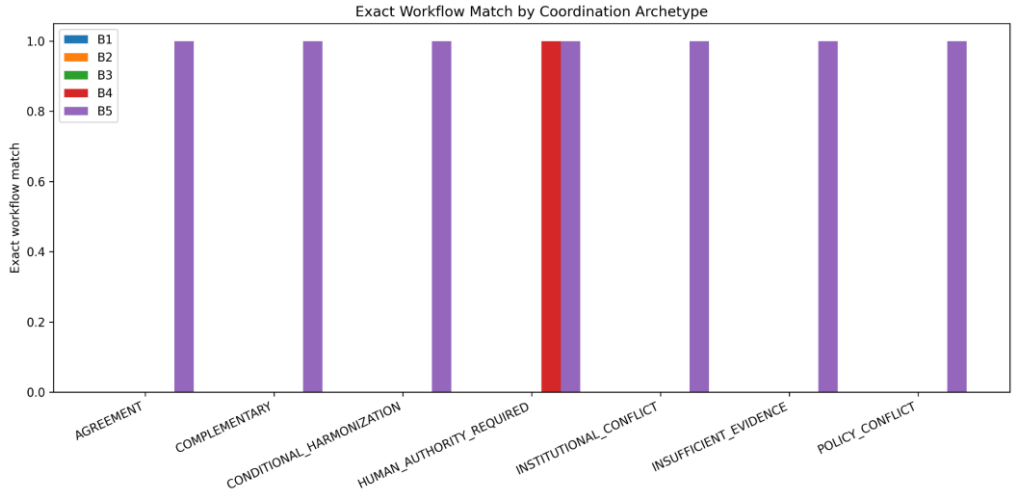


Figure D3. B5 matches all seven encoded archetypes; B4 exactly matches the human-authority archetype. Ten variants per archetype do not create ten independent institutional problems.

Table D1. Baseline architecture definitions and boundaries.

Baseline	Frozen implementation / access	Interpretation
B1	All available information; valid-looking actions issued under omniscient_agent ownership.	Ownership is considered unauthorized in this benchmark. This is not a theorem against centralized planning.
B2	Role-named proposals are emitted without enforced policy/mandate boundaries; modeled evidence broadcast.	Isolates the insufficiency of role labels alone.
B3	Majority-intent vote; ties escalate; evidence summaries broadcast.	Retains the submitted consensus/debate name with an implementation qualifier. No live debate model tested.
B4	Independent policy/mandate/evidence filtering; unresolved cross-role cases may escalate.	Not the same method as Gate 29's capable stateful selector.
B5	Structured-fact harmonization with allowed alternatives, messages, conditions and escalation.	Reference-rule consistency. Subsequent tests isolate scheduling, choice and timing separately.

B1/B2 overexposure is scored against modeled required-field lists, not measured access to actual patient records. Architectural-reference runtime is in-memory reference-logic timing; it must not be compared directly to later HTTP latency or called LLM response time.

E1. Dependency-aware proposal scheduling

Only proposal scheduling changes; institutional observations, proposal ownership and validation remain fixed. Eight known workflow families yield 318 distinct family–schedule pairs per method, or 1,590 method executions. The figure uses equal-weight macro-family completion. [9] Figure E1 reports order-robust completion, and Figure E2 compares validator-call efficiency.

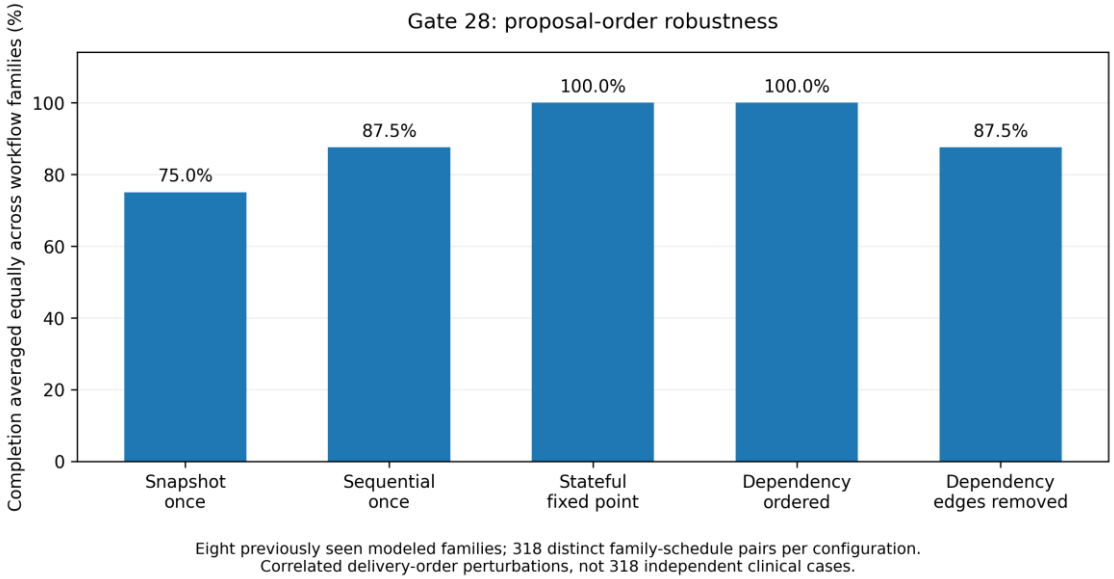


Figure E1. Snapshot once 75.0%; sequential once 87.5%; fair retry and dependency ordering 100%; removed edges 87.5%. The strong-control tie is retained.

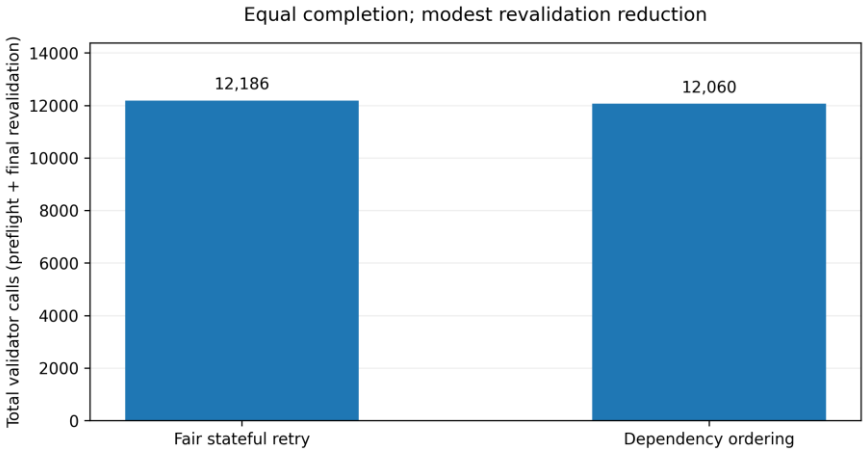


Figure E2. At identical completion, dependency ordering uses 12,060 validation calls versus 12,186: 126 fewer (1.03%). Operation counts do not establish a latency speedup.

Removing only dependency edges reproduces sequential-once effects in all 318 paired comparisons. This isolates a scheduling effect. It neither changes B1–B5 nor tests negotiation over previously unavailable alternatives.

E2. Authorized workflow selection

The authorized-workflow study compares choice mechanisms under the same contracts and objective. Four development and 14 internal confirmation families are used; ten confirmation families admit a complete workflow, while four require partial work, additional evidence or authority handling. [9] Figure E3 compares complete-workflow recovery, and Figure E4 reports agreement with the best permissible complete or partial plan.

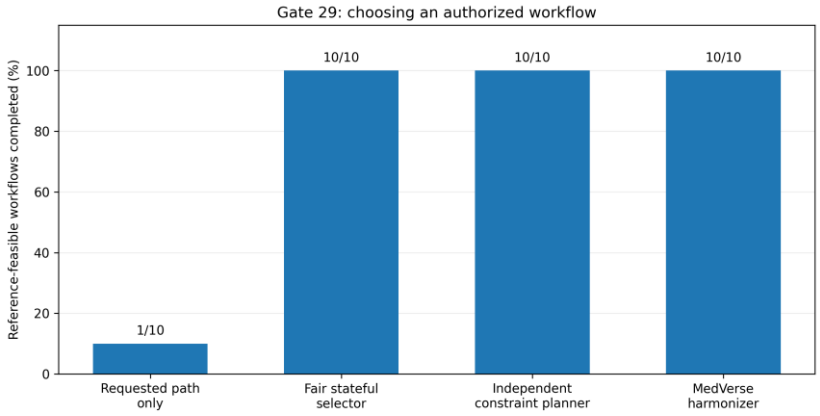


Figure E3. Requested-path-only completes 1/10; fair stateful selection, independent constraint planning and MedVerse each complete 10/10. The requested-path control is intentionally limited, not the strongest comparator.

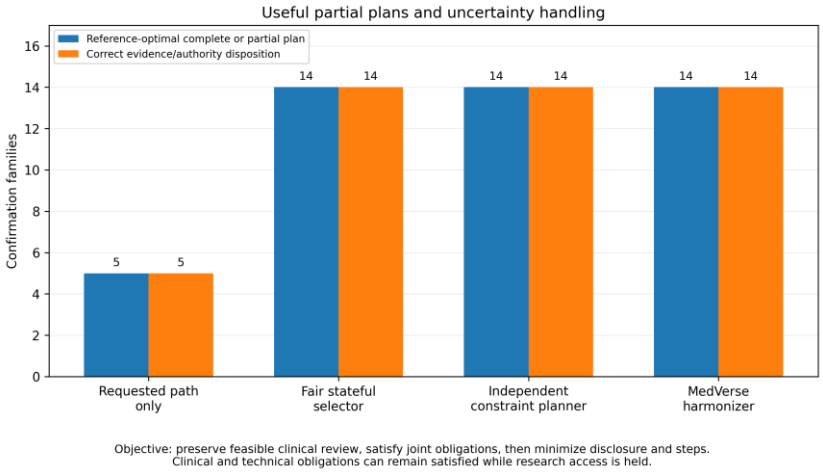


Figure E4. The capable methods agree with the best permissible complete/partial outcome and disposition on 14/14 families. Equally valid workflows are accepted.

The objective preserves feasible clinical review, then joint obligations, then minimizes disclosed field–recipient pairs and action count. Hard authorization constraints cannot be traded away for a higher utility score. These snapshot results are not end-to-end progressive guarantees.

E3. Policy-context handoffs and progressive execution

Figure E5 reports receiving-side context validation, and Figure E6 shows the remaining snapshot-to-progressive execution gap.

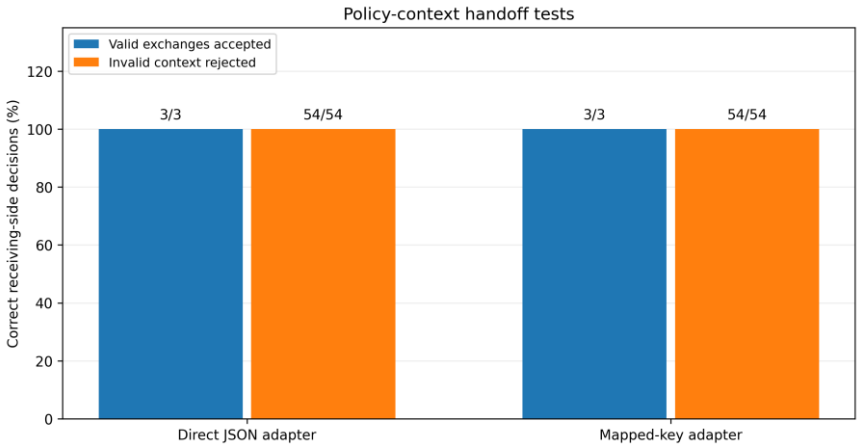


Figure E5. Two local adapters: six valid exchanges accepted, 108 altered/incomplete contexts rejected. Correlated probes of three base contexts; not authenticated remote interoperability or complete ITU conformance.zzzzzzzzzzzz

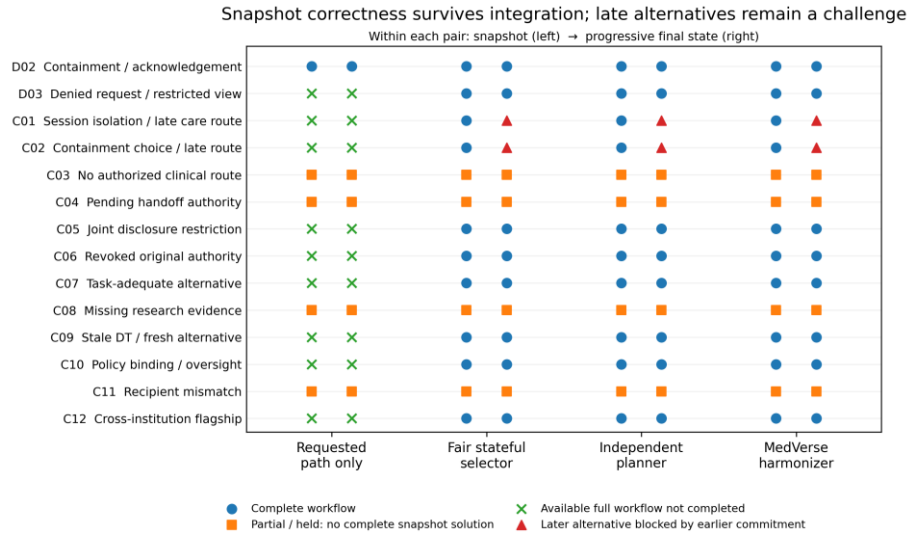


Figure E6. Snapshot-to-progressive comparison on known families. Capable methods fall from 10/10 feasible snapshots to 8/10 progressive completion because of early irreversible commitments in C01/C02.

Those progressive failures remain in the release. The later commitment study introduces explicitly declared timing contracts in a separate benchmark; it does not retrospectively insert future information or rewrite C01/C02.

F1. Digital Twin provenance and permitted evidence scope

Figure F1 shows the field-specific freshness diagnostic, and Figure F2 reports the role-eligible evidence scope.

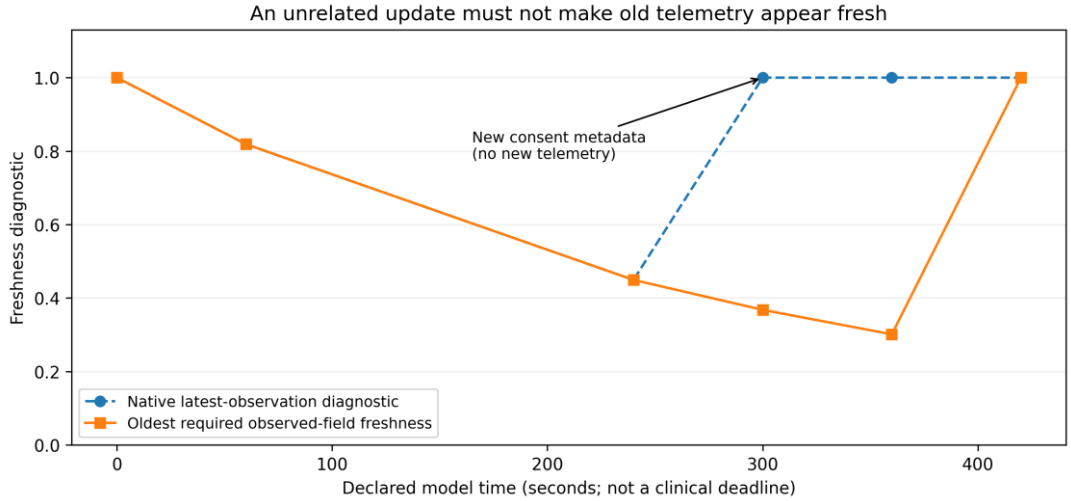


Figure F1. New consent metadata must not make old telemetry look fresh. The adapter tracks required observed fields separately; native historical rules remain unchanged.

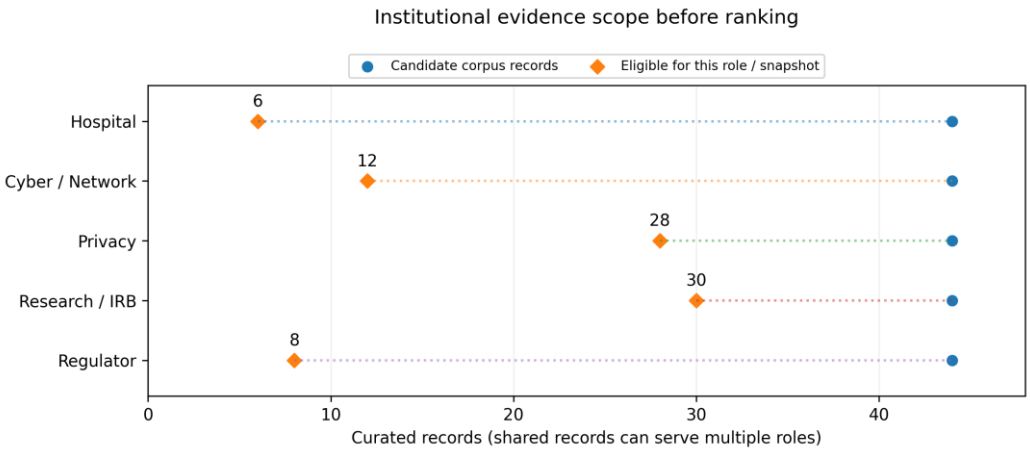


Figure F2. Role-eligible records: Hospital 6; Cyber 12; Privacy 28; Research 30; Regulator 8. Shared records can serve multiple institutions. Counts are not corpus completeness or agent accuracy.

PatientDigitalTwin, DigitalTwinQuality and ProvenanceRecord are project-local Python classes. The observation adapter does not turn demonstration defaults or transferred-envelope metadata into patient observations. DT validity is separate from action permission.

F2. Korean/English policy retrieval

Figure F3 shows paired Korean/English BM25 target-source ranks, and Table F1 compares the aggregate lexical and dense retrieval results.

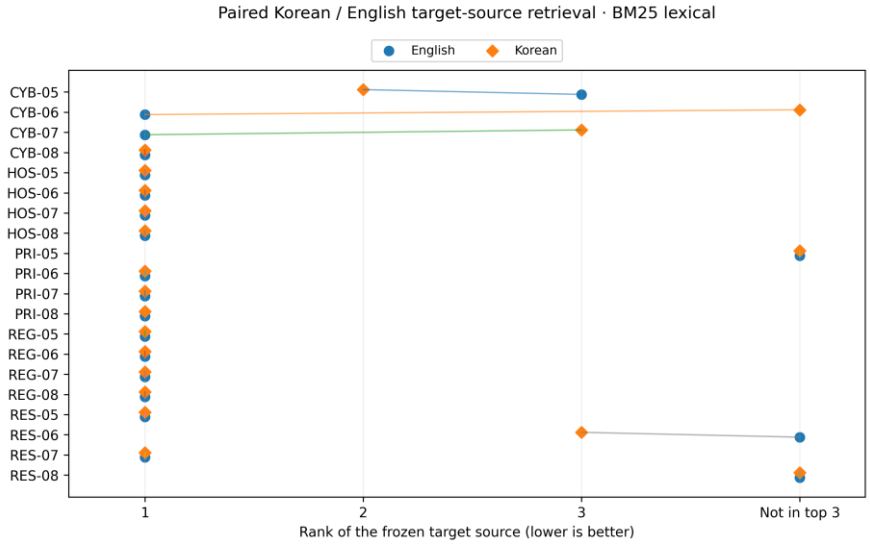


Figure F3. BM25 paired target-source ranks on 20 previously evaluated intent families. Aggregate recall equality does not imply identical question-level outcomes.

Table F1. Korean/English policy-retrieval results.

Backend / evidence origin	EN top-1	EN Recall@3	KO top-1	KO Recall@3	Both languages hit
BM25 / archived regression	16/20	17/20	14/20	17/20	16/20 families
Dense / supplied Windows console	13/20	19/20	10/20	15/20	15/20 families

The lexical and dense backends were evaluated on the same predefined target-source questions. Dense retrieval improves English Recall@3 but underperforms BM25 on Korean Recall@3; neither backend dominates across both languages. These are known-query retrieval regressions, not legal-applicability validation. Run identifiers and environment details are retained in Annex I1.

Dense retrieval uses sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 with normalized 384-dimensional vectors and top-k = 3. Model revision and environment provenance are listed in Annex I1.

The local software environment and model-version observations are recorded for reproducibility in Annex I1. Official source text remains authoritative over translations or paraphrases.

F3. Progressive workflow trace and runtime measurements

Figure F4 shows the progressive clinical/research execution trace, and Figure F5 reports the corresponding local HTTP response distribution.

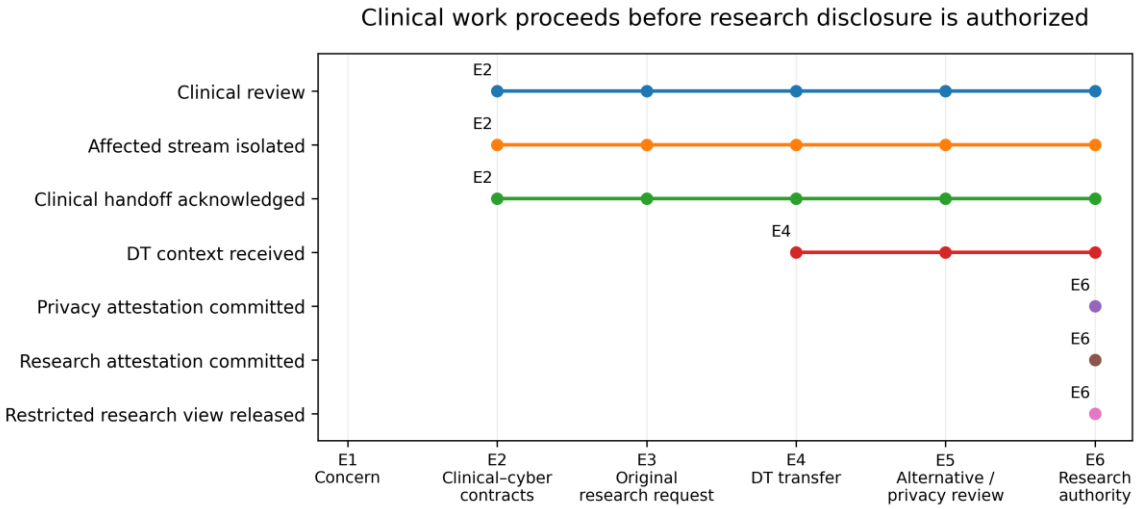


Figure F4. Actual C12 HTTP trace: modeled clinical review at E2; restricted research view at E6. The two permissions are not interchangeable.

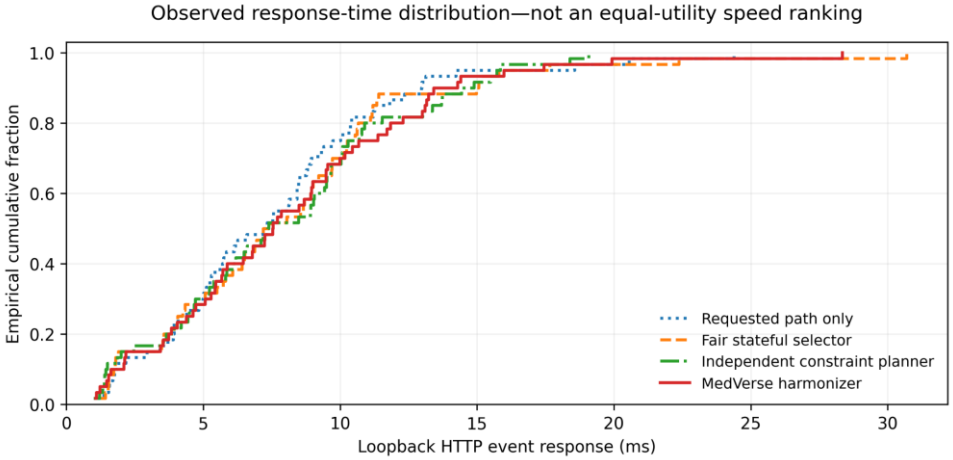


Figure F5. Progressive-integration HTTP response ECDF: 60 event calls per configuration, ten C12 replays, warm-up excluded and randomized configuration order.

Local loopback measurements confirm interactive execution but exclude external networking, model inference and human review. They are not used as a speed-superiority claim; detailed runtime provenance is retained in Annex I1.

G1. Bounded commitment under partial information

The selector and validator are held fixed while commitment timing varies. Four development prefix families / eight branches precede eight internal confirmation prefix families / 32 branches. Identical visible histories must yield identical deterministic decisions. [9] Figure G1 summarizes family-level completion, and Figure G2 shows the completion-versus-waiting trade-off.

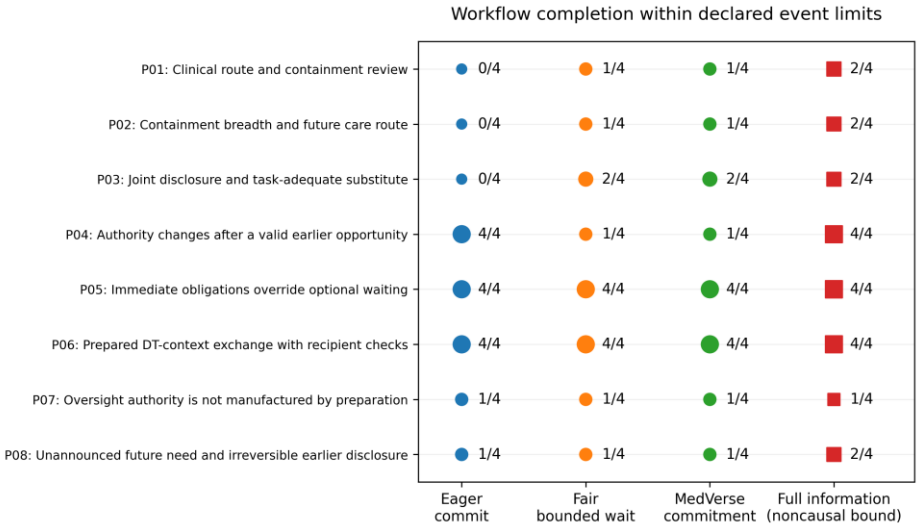


Figure G1. Family-level outcome map. Eager 14/32; fair bounded wait and MedVerse 15/32. The full-information upper bound is noncausal and not a deployable comparator.

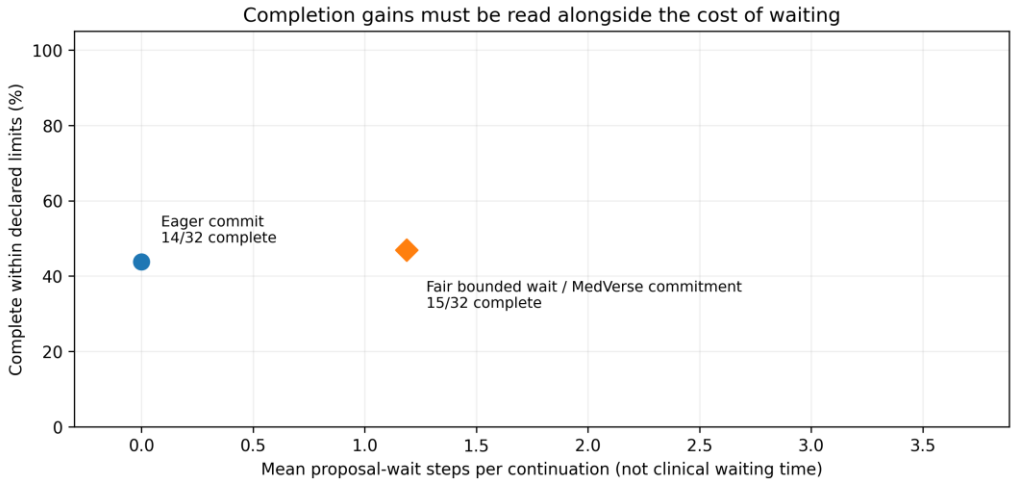


Figure G2. Four gains and three losses versus eager commitment; capable controls tie on all 32 branches. Mean proposal-wait steps 1.19, not patient waiting time.

A validated action today may lose its opportunity if authority changes during a permitted wait. Later revocation does not erase earlier valid effects. Bounded waiting is therefore a visible institution-owned contract, not a global promise of improved care.

G2. Institutional contributions and commitment trace

Figure G3 makes commitment timing inspectable, and Figure G4 reports institution-specific contributions.

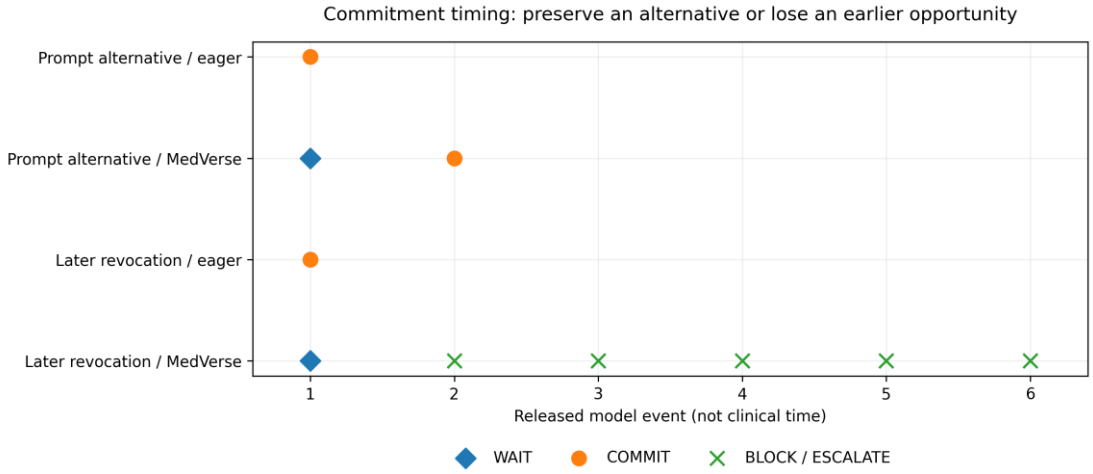


Figure G3. A prompt alternative can justify waiting; later revocation can remove an opportunity. COMMIT symbols are actual receipts, and each lane ends after its obligation is achieved.

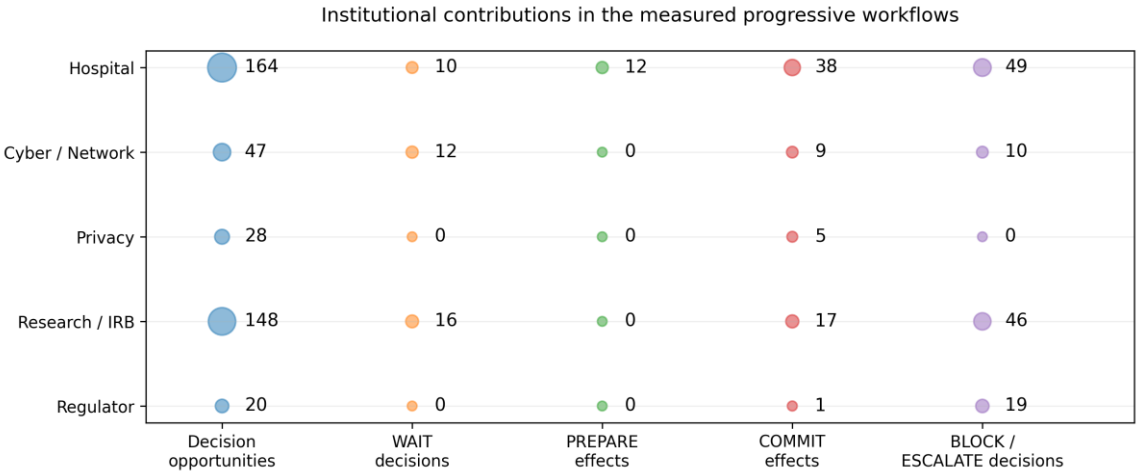


Figure G4. Workspace institution-level behavior. Hospital 164 opportunities / 38 commits; Cyber 47/9; Privacy 28/5; Research 148/17; Regulator 20/1. Opportunities are reconsidered events, not independent predictions.

This addresses agent visibility without claiming every agent is a trained classifier. C12 legitimately has no Regulator-owned proposal; oversight is shown in P07. The operator inspector is not a proof of authenticated inter-institution communication or production isolation.

G3. Interface integration and execution fidelity

Figure G5 reports the operator-response distribution, while Table G1 verifies agreement between the frozen engine and HTTP/UI execution.

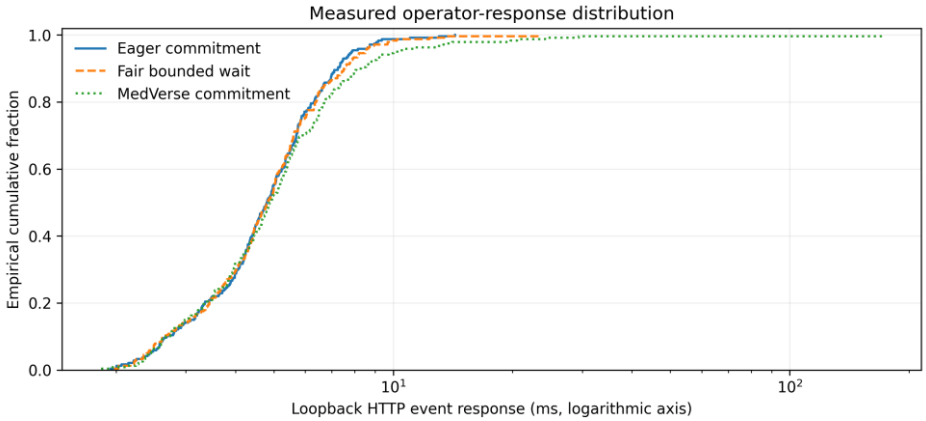


Figure G5. Workspace response distribution, logarithmic time axis. 240 event calls per controller; one known-case replay without randomized repeated trials. No speed-superiority claim.

Table G1. Interface-integration checks.

Check	Recorded result	What it establishes
Wrapper/input/role checks	47/47	Defined engineering checks pass; not an external security certification.
Engine versus HTTP event state	720/720 identical	The interface retains the frozen controller's behavior.
Exported traces	120/120 identical	Trace export matches the actual evaluated execution.
Identical-history comparisons	177/177	No measured use of hidden continuation information in these probes.
Changed-authority probes	72/72 no new unauthorized effect	Final revalidation blocks affected commitment in the tested perturbations.
Earlier clinical outcomes	432/432 unchanged	The new workspace did not rewrite the earlier clinical flow.

The commitment and interface-integration studies repeat some invariants and cases; their check totals are therefore not additive evidence. Native-machine and browser-download verification remain deployment checks rather than new scientific samples.

The commitment and original clinical/DT workspaces share a host but remain separate modeled sessions. Candidate action contracts, review records and effects are not real hospital signatures, real research approvals or a connected clinical service. Clinical/legal interpretation review remains pending.

H1. Cross-domain reuse: same coordination mechanisms, different domain contracts

Table H1 maps reusable coordination functions from the medical-metaverse configuration to a possible education-domain configuration.

Table H1. Cross-domain reuse mapping.

Reusable function	Medical-metaverse configuration	Possible education configuration
Domain service	Hospital	School / educational-service provider
Technical integrity	Cyber/Network	Platform / assessment-data integrity
Data protection	Privacy	Learner-data protection
Research governance	Research/IRB	Applicable education-research ethics
Oversight	Regulator simulation	Applicable education oversight simulation

A valid transfer would need new authoritative sources, observations, purposes, mandates, action contracts, authority records, task requirements and independently specified outcomes while preserving the reusable engine. No education performance result is included because none was established in the supplied evidence.

H2. Detailed mapping of observed requirements to standards and policy gaps

Table H2 links the observed coordination requirements to proposed domain contributions and their remaining standards or policy boundaries.

Table H2. Detailed standards and policy-gap mapping.

Observed requirement	Proposed domain contribution	Mapping / remaining boundary

Clinical–cyber responsibility handoff	An owned, acknowledged handoff that records affected evidence, care obligations and permitted routes.	D7/D8; WHO accountability [5]. Clinical plausibility and timing contracts require expert review.
Purpose-specific shared-workspace access	Bind purpose, recipient, fields, task adequacy and current authority to each proposed alternative.	D10; OECD health-data governance [6]. A smaller view is not automatically lawful; Korean applicability remains pending.
Policy-aware DT exchange and commitment	Carry provenance, freshness, permission scope and validity conditions; recheck before any irreversible effect.	Y.4239/Y.4240 [3,4] already cover access and exchange functions. A healthcare profile is proposed, not a proven omission or full-conformance result.

I1. Evidence provenance, figure register and reproducibility

No experiment is rerun by this revision. Figures are reused from the supplied attachments or preserved release archive. Existing result values and known limitations are retained. Any source-summary correction is documented below rather than altering historical artifacts. Table I1 records the evidence artifact, run identifier and interpretation limit for each reported result.

Table I1. Evidence provenance and reproducibility register.

Evidence	Identifier / artifact	Do not overstate
Learning	outputs/gate24c/gate24c_status.json; gate24c_action_class_metrics.csv; gate24c_hyperparameter_results.csv; gate22b6_status.json	Cyber 54 TRAIN / 18 DEV / 72 prospective; Privacy 486 fresh contexts over finite states.
Original B1–B5	outputs/gate26/gate26_status.json and original comparison runner; supplied five-page report and figures	Non-timing values agree with supplied local log. Runtime varies by run; no combined latency estimate.
Scheduling	Gate 28 report: 20261103T035936_964985Z	318 effective pairs, eight known families, five configurations.
Alternative choice	Gate 29: 20261103T075734_173247Z	4 development / 14 internal confirmation families; 72 executions.
Progressive integration	Gate 30: 20261103T111811_756965Z	14 evaluated families; 10 snapshot-feasible; C01/C02 retained.
DT / BM25 bridge	Gate 31: 20261103T132213_435523Z	Dense dependency blocked in the attached PDF. Historical BM25 replays only.
Dense local evidence	User console: 20261104T003737_462827Z	Successful pinned-constructor CPU run; user-supplied summary, not rerun here.
Commitment	Gate 32: 20261104T013623_796085Z	8 confirmation prefixes / 32 continuations; comparison with hindsight bound.
Workspace integration	Gate 33: 20261104T035439_172645Z	Known-input HTTP replay; no controller retuning.

I2. References, source links and supplemental file map

[1] ITU-T Y.3172 (2019), Architectural framework for machine learning in future networks including IMT-2020.Official source

[2] ITU, AI Ready — Analysis Towards a Standardized Readiness Framework, Report 2.0, January 2026; §3 Dimensions.Official source

[3] ITU-T Y.4239 (2025), Reference model for integrating virtual and physical worlds through digital twins in the metaverse.Official source

[4] ITU-T Y.4240 (2025), Interoperability for integrating virtual and physical worlds through digital twins in the metaverse.Official source

[5] WHO, Ethics and governance of artificial intelligence for health (2021).Official source

[6] OECD, Recommendation of the Council on Health Data Governance, OECD/LEGAL/0433.Official source

[7] Team MedVerse candidate Korean corpus v0.4 and source snapshots. Official Korean originals are the policy anchors; interpretation review remains pending.

[8] Sentence Transformers, paraphrase-multilingual-MiniLM-L12-v2 model card.Official source

[9] Team MedVerse frozen Gates 22B.6–33, supplied local console results, status files and source implementations; identifiers listed in I1.

[10] AI for Good Sandbox Hackathon submission template, supplied by the user; UAE competition requirements.Official source

Official source[11] ITU-T Y.3060 (2023) — Autonomous networks: Overview on trust.

[12] ITU-T Y.3061 (2023) — Autonomous networks: Architecture framework.

[13] ITU-T Y.3090 (2022) — Digital twin network: Requirements and architecture.

[14] National Cancer Center, Republic of Korea — Cancer Incidence Statistics (Public Data Portal, 2026).

[15] ITU-T Focus Group on Metaverse (FG-MV) — approved deliverables on metaverse architecture, cyber risk, PII and interoperability.

[16] ITU/WHO Focus Group on AI for Health (FG-AI4H) — deliverables on ethics, regulation, data, evaluation and AI lifecycle.

External material informs standards and governance comparison. It is not substituted for a Korean action's lawful basis. This revision does not newly certify the completeness of the policy corpus or correctness of each operational interpretation.

J1. Patient Digital Twin modeling and source-to-state trace

This note clarifies the existing implementation; it does not introduce new experiments, change thresholds, or claim a live patient connection.

Intended representation versus current implementation

The patient Digital Twin is intended to represent the available observations of the patient's evolving condition, together with the uncertainty and governance context attached to those observations. In this release, the implemented state is a latest-observation model: individual fields are updated from explicitly supplied scenario observations. It is not a physiological simulator, calibrated disease-progression model, raw biosignal processor or validated diagnostic system. Table J1 traces how observations enter the Digital Twin, retain provenance and remain separated from authorization.

Table J1. Patient Digital Twin source-to-state trace.

Step	Implemented behavior and relevant source
Receive a reading	medverse/digital_twin/model.py updates the named clinical/network/security field and appends a ProvenanceRecord. Original S1 numeric clinical fields are SpO2 and heart_rate.
Preserve its context	foundation31/digital_twin.py starts with empty observed fields, records observation/receipt time, rejects future/replayed/stale-field input, and retains role-scoped views.
Evaluate evidence quality	The adapter checks missing required fields, oldest-field freshness and the original integrity conditions. A new consent record does not freshen older telemetry. These are diagnostics, not clinical timing standards.
Produce a reference assessment	Historical Hospital modules use numeric observations to propose measurement verification, contact and human clinical review. Their frozen reference thresholds differ; they are not one clinically calibrated decision model.
Authorize a workflow	Institution-owned proposals remain subject to mandate, purpose, evidence and current-authority checks. A clinical trigger, a DT-quality label or a retrieved policy passage does not itself authorize execution.

Important live-path boundary

The current foundation adapter projects released role context and receiving-envelope metadata; it does not create unprovided SpO2, heart-rate or temperature measurements. The commitment workspace uses institution-owned contracts in a separate modeled session. The interface therefore represents a controlled workflow prototype, not autonomous examination of a real patient from live sensors.

The original observation module and reference reasoner can be replayed for explanation. For the existing S1-like values, missing core metadata and later identity failure keep the adapter's evidence state UNVERIFIED while the historical Hospital reference still proposes measurement verification and clinical review. This illustrates separation of clinical concern from evidence trust, not a validated treatment decision.

J2. Korean open-data provenance and demonstration plan

The original source was not deleted from the design

config/data_manifest.json retains KR_NCC_CANCER_INCIDENCE_20260120 and medverse/data_grounding/ncc_context.py retains its CSV adapter. The official record is National Cancer Center Cancer Incidence Statistics, public-data identifier 3039563. The provider describes aggregate statistics for 1999–2023, not patient-level sensor trajectories. [12] Table J2 summarizes the dataset provenance, current integration status and restoration requirements.

Table J2. NCC open-data integration status.

Item	Status in the inspected RC1 archive
Raw data	data/external/ncc_cancer_incidence.csv is absent; .gitignore excludes external CSV files by default.
Legacy integration	Gate19 optionally loads the CSV and attaches descriptive context to Hospital/Research evidence bundles; clinical_inference_permitted remains false.

Current live integration	No NCC loader call appears in the Gate30–33 runtime. Retained source code is not evidence that the GUI currently uses the dataset.
Appropriate role	Population background, cohort-description context and motivation. Never substitute prevalence/incidence for an individual measurement, diagnosis, risk estimate or clinical trigger.
Restoration dependency	Download the official file, validate schema/version/hash and expose a separately labelled context view. Inspect category overlap and denominators before interpreting aggregate counts or crude-rate summaries.

A clearer demonstration structure

Use short opening slides to explain the problem, sensor-to-state representation and the three distinct evidence layers. Show the actual launch command, open the original DT workspace, and distinguish observed context from a full sensor-processing pipeline. Then demonstrate P01/PROMPT, P04/NO_REPLACEMENT, P07/APPROVAL and changed-authority rejection. Show actual GUI tables and the source-labelled archived plot gallery, then close with implications and review boundaries. The GitHub publication tutorial belongs in a separate handover guide, not the seven-minute competition recording.

Publication and provenance

The presentation companion is read-only and additive; archived figures do not change with the current session. Report source runs and the active backend explicitly. Code publication must retain required freeze artifacts, exclude credentials and unapproved data, respect source-document rights, and pass a clean-clone reproduction check. No scientific results were revised in this document update.

[14] National Cancer Center, Republic of Korea. Cancer Incidence Statistics (국립암센터_암발생 통계 정보), data.go.kr record 3039563, source metadata updated 20 January 2026. https://www.data.go.kr/data/3039563/fileData.do. Metadata checked 5 November 2026; raw CSV not ingested in this editorial update.

Implementation evidence: Team_MedVerse_DT_Data_Implementation_Audit.md; source paths and hashes, optional-input status, and the explanatory replay are supplied with the presentation update.